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In-situ sedimentary evidence of complex bottom currents at a modern deepwater seamount

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Interactions between deep-ocean currents and seafloor sediments create bedforms that are important for paleoenvironment reconstructions, resource exploration, and mapping ecosystems. Oceanographers recognize complex flow dynamics where deep ocean currents interact with submarine topography (i.e. seamounts) however the effects on below-wave base seafloor sediment remain unquantified. Here we show complex, highly variable bedforms in the sediments around a seamount are linked to deep-ocean current and topographic interactions, using in-situ, high-resolution footage from a <10 year-old, 1-km water depth volcanic seafloor. Bedforms record sediment reworking, vary spatially and temporally at the scale of m – 100 s m and reflect changes in current velocity over hours to days. These observations provide insights into the complex sedimentology associated with current-topography interactions in mid- to deep-ocean settings and demonstrates the limitations of existing sedimentological models. This study highlights that primary seafloor data can be quickly obscured, impeding modern data collection, subsequent hazard modelling, and paleoenvironment reconstruction.

Deep-ocean currents play an important role in seafloor morphology, ecosystems, and global cycles¹. These currents can reach speeds in the order of 1 ms⁻¹, remobilize sediment, disperse organisms, microplastics and seafloor sulphides, and affect benthic habitats². Nearly 60 years ago, the first contour-hugging bottom currents were identified in the north Atlantic from deep-sea photographs of current ripples³. Since then, significant work has focused on deep-sea deposits such as contourites^{4–7} and turbidites^{8–11}, however very little work has focused on the sedimentary response to currents around large-scale topographic features in the open ocean such as seamounts, knolls, and abyssal hills. Considerable oceanographic work around seamounts (topographic features that rise 1000 m or more from the seafloor but do not penetrate the sea surface) has demonstrated complex and unusual circulation effects^{12–21}, with currents being spatially and temporally variable. Numerical modelling of current-seamount interactions^{19–22} and in-situ studies^{23–26} have shown that bottom currents around seamounts (including jets, eddies, circulation cells) can reach speeds sufficient to mobilize sediment, and have time-scale fluctuations on the order of days. Indeed, variability in sediments around seamounts has been reported^{14,16,27,28} although these studies did not have sufficient scope for a detailed sedimentological investigation. As a result, how these complex and highly changeable bottom currents affect seafloor sediment remains an open question that has not been properly captured in sedimentological models²⁹.

Understanding seafloor processes can also provide insights into the impact on benthic communities. Disturbance regimes are characterized by areas of the seafloor that are subject to episodic disruptions (e.g. current scouring, chemical or biological impacts) at an interval roughly matching the benthic community recovery time³⁰. Recurring disturbances can result in a patchy landscape exhibiting a range of stages of colonization from barren to fully mature, containing a greater biodiversity per unit area than a disturbed or undisturbed habitat alone^{30,31}. As marine conservation initiatives require proper mapping and documentation of ecosystem processes, understanding the influence of bottom currents as drivers of benthic disturbances is key.

This study targets the remobilisation of fine-grained, silicic products (rhyolitic ash) of the 2012 deep underwater eruption of Havre Seamount, a submarine silicic volcano in the Kermadec arc/Rangitāhua³². The Havre Seamount is approximately 25 km in diameter with a 5 km wide caldera. The caldera rim sits at 900 meters below sea level (mbsl) and the surrounding volcanic flanks extend down to 1750 mbsl, with a complex morphology consisting of lava domes, lineaments, hills, and arcuate ridges³³. The seamount was visited twice by our team since its 2012 eruption. The present study is based on a compilation of datasets from the MESH voyage in 2015 on the RV *Roger Revelle*, consisting of video footage from the remotely operated vehicle (ROV) *Jason*, a high-resolution bathymetry (1 m)

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from the automated underwater vehicle (AUV) *Sentry*, and a gravity core, as well as the 2022 VULKA expedition on the CSIRO research vessel *RV Investigator* that gathered additional video footage using a deep-towed camera (see: 'Methods'^{32,35}). Complimentary data of regional mid-ocean current velocity was acquired for ARGO floats from the Scripps Institute of Oceanography data collection³⁶. This paper is based on 433 unique seafloor bedform measured on and around the seamount, from 794 mbsl (meters below sea level) on the caldera rim to 1662 mbsl in the surrounding basin.

The Havre Seamount is part of the Tonga-Kermadec arc/Rangitāhua that extends for 1200 km in the southwest Pacific and links Tonga to New Zealand. This intra-oceanic arc is volcanically very active and consists of dominantly submarine volcanoes with only a few emergent islands³⁷. Silicic explosive activity is widespread and both subaerial and submarine activity have been documented^{38–40}. The Havre Seamount sits at 31° 05'S 179° 5'W and summits as a 5-km-wide, 900 mbsl caldera rim (Fig. 1). The seamount flanks extend down to 2400 mbsl with a slope of ca. 3° (Fig. 1B). Seafloor footage on and around Havre Seamount features thick accumulations of 2012 pumice lapilli (grainsize: 2–64 mm) to boulders (up to 10 m), which are topped with laterally extensive ash (<2 mm) layers^{32,35,41}.

Here, we present a high-resolution dataset which shows evidence that the deep-sea bedforms at Havre Seamount are remobilized from spatially and temporally shifting ocean currents that affect the seafloor. Observed bedforms suggest extreme process variability at the local (m to km) scale driven by the topographic effects of the seamount and its secondary features on impinging currents. The observed variability provides insights into the nature of deep-sea sedimentation, beyond existing models that focus on processes associated with continental margins (i.e. contourites, turbidites).

The observed bedforms at Havre change in size, orientation, and nature over m-scale distances and in yearly (or less) timescales and can be partly correlated with local current effects. We show evidence that considerable areas of the deep seafloor are likely to experience pronounced and variable sediment remobilisation beyond what has been previously reported, and introduce a new facies model, *Deep-water Multidirectional Tractional Sands* (DMTS), to address this gap. We discuss differences between DTMS and other established facies models and demonstrate that interpretations of seafloor sediments can be challenging and potentially incomplete or misleading. Further, we extend our model to ancient environments and interpretations made from sedimentary structures (paleocurrent indicators) in the rock record (i.e. outcrop, core), which are commonly used for basin

interpretation and resource exploration. Our model also shows complex bottom currents may be an impediment to modern data collection and the modelling of volcanic events, which has implications for hazard assessment, and that complex bottom currents may impact benthic communities and biodiversity.

Results

The 2012 submarine eruption at Havre Seamount produced rhyolitic ash that covered the entirety of the submarine caldera and its outer flanks^{32,41}. Interestingly, our 2015 and 2022 surveys show that ripples and scour marks are ubiquitous in the widespread 2012 ash and lapilli and older deposits, demonstrating that deep ocean bottom currents are active in reworking the material in the months after the eruption. Our initial survey was less than 3 years post eruption and showed well-developed, complex ripple fields. Erosional and depositional bedforms occur concurrently on the seafloor suggesting considerable variability in current velocity (Fig. 2). Ripple marks are the dominant sedimentary structure around the caldera, followed by obstacle-and-scours, occasional dunes, and rare sand ribbons and sand furrows. Full dataset of observations and measurements available in Supplementary Data.

Ripples and obstacle-and-scours

Sediment ripples occur in localized patches on the rim and floor of the caldera, and as more widespread sections on the outer caldera flanks. Ripples occur on the outer slopes as far as 10 km from the caldera rim of Havre Seamount but diminish in frequency; they are absent in the surrounding basins (ca. 2000 mbsl, Fig. 2A). Three ripple types are identified in the study area (Supplementary Movies 1 and 2). Type 1 ripples or topographically bound ripples are localized and associated with bottom current interaction with small to intermediate-scale topographic features (m–100 m size, e.g. lava domes, boulders, channels, steep slopes, Fig. 3A–C). Ripples may change direction, size and shape depending on a change or break in slope, may deflect around an obstacle, or reflect the acceleration of currents within channels or downslope. Type 2 ripples or slope ripples are more widespread (10 s–100 s of m) and form bedform fields along the caldera slope, where there are fewer topographic features (Fig. 3D). Slope ripples do not exhibit a consistent orientation across the study area and may change direction abruptly, with no discernible cause (Fig. 4). Seemingly random changes in slope ripple orientation without any apparent influence suggests that the

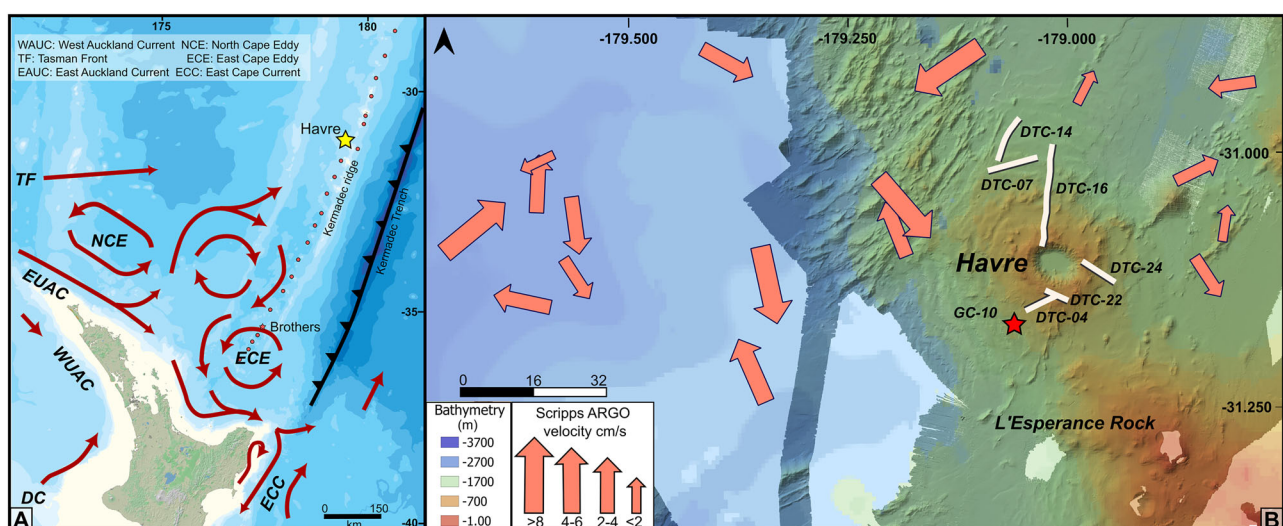


Fig. 1 | Location of the Havre Seamount. Regional and local setting with ocean currents. A Surface currents north of New Zealand and the location of the Kermadec Ridge, with Havre Seamount (yellow star) and adjacent seamounts (red dots). B DEM of Havre Seamount and nearby topography (L'Esperance Rock), with

Scripps ARGO velocity product (pink arrows) from 2007–2020 at 1000 mbsl, and deep-towed camera transect lines (white lines). Red star indicates the location of the CG-10 Core. Background bathymetry provided by GEBCO (Derived product from the GEBCO 2021 Grid, made with NaturalEarth by OpenDEM).

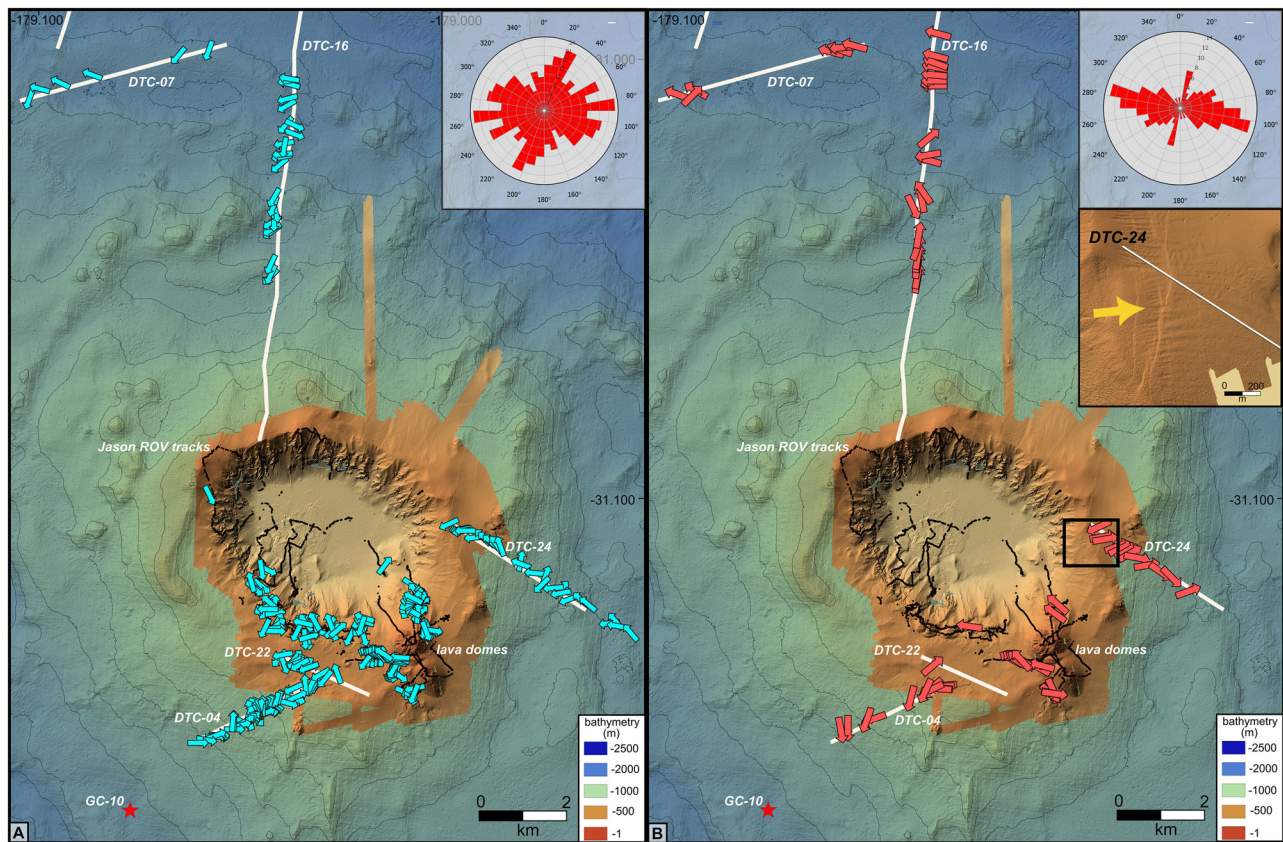


Fig. 2 | Interpreted current directions from observed bedforms around Havre Seamount with 1 m DEM of Havre caldera (highlighted topography).

A Interpreted current direction from ripples (blue arrows) along Deep-Towed Camera lines (white lines) and on ROV tracks (black dots) and associated rose diagram of ripple orientation. **B** Interpreted current direction from scours (red

arrows) and associated rose diagram. Inset map: AUV Sentry DEM of the eastern seamount rim. Linear features are interpreted to be deep scours where current flow is accelerated down the seamount flanks. Background bathymetry provided by GEBCO (Derived product from the GEBCO 2021 Grid, made with NaturalEarth by OpenDEM).

bottom currents also experience sudden directional changes. Type 3 ripples or reworked ripples occur locally (ca. 10 s of m) as ripples and sand ribbons that interfinger at contrasting angles (Fig. 3E, F). Reworked ripples indicate a change in bottom current velocity, where previously deposited ripples are reshaped under a new current direction into sand ribbons.

Obstacle-and-scours occur in the silt- to sand sized seafloor sediments around larger clasts (pebbles to boulders). These bedforms are a few cm to dm deep and have an area roughly scaling with the size of the clast (Supplementary Movies 1 and 2; Fig. 3G). Spatially, obstacle-and-scour orientations are more consistent than observed ripple orientations, although still exhibit a wide range of direction across the study area (Fig. 2B). Rare obstacle-and-scour features are contrary (i.e., 180°) to the dominant direction, suggesting changes in current velocity or extremely local (10 m) eddying. Observed orientations also suggest the deflection of currents around small- to intermediate scale topography (similar to Type 1 ripples) as given by the obstacles-and-scours around the lava domes on the south-eastern flank of the seamount (Fig. 2B). A field of larger-scale, deep scour marks (100 s m) in fine- to very fine-grained sediment (as indicated in transect DTC-24) immediately to the northeast of the caldera rim is visible from the 1 m DEM obtained by AUV Sentry (Fig. 2B, inset⁴²). These larger features are 5–10 m wide and 100–200 m long, in areas of steep slopes (10–20°) immediately outside and radiating from the caldera rim. Occasionally, obstacle-and-scours and ripples are observed together on the seafloor, with ripples in the sediments around the scoured surface. On these instances, obstacle-and-scour orientation and ripple orientation are at contrary angles (Fig. 3H, I).

The structures around Havre Seamount indicate that there is considerable reworking of the seafloor sediments by irregular bottom

currents in the years following deposition. Ripple and obstacle-and-scour orientations are variable at all locations around the seamount, suggesting the lack of an overarching and persistent hydrodynamic regime. Additionally, observations on the seafloor suggest active variations in the currents within the 3- and 10-year timescales of observation. We interpret Type 1 ripples (topographic ripples) as being governed by small-scale (m–10 s m) local topography that serves to steer or enhance local bottom currents. Type 2 ripples (slope ripples) represent more widespread bottom current flow, however abrupt changes in direction demonstrate these currents are not persistent and vary. We suggest that current velocity changes arise from impinging or competing currents that serve to steer or alter the existing current, which may arise from more regional variability, local eddies, incident jets, or density interfaces creating turbulence^{21,23}. Type 3 ripples (reworked ripples) are interpreted as the result of changes in current velocity where one current is superseded by another. Type 3 ripples form under a bottom current flowing in one direction, which is followed by a change in current velocity that reworks the ripples into sand ribbons aligned under a new current direction. Concurrent obstacle-and-scours and ripples with contrasting orientations occur indicate the current direction given by the obstacle-and-scour is likely the prevailing direction for the area, while the ripples surrounding the feature represent more short-term fluctuations in current velocity. Type 2 and 3 ripples, and contemporaneous obstacle-and-scour marks and ripples with conflicting orientations suggest that changes in current velocity are frequent and occur at minimum within the 3- and 10 years of observation. The large-scale scours on the eastern caldera rim are concluded to be the result of long-duration, accelerated current flow (ca. 1.5 ms⁻¹) down the outside rim of the caldera.

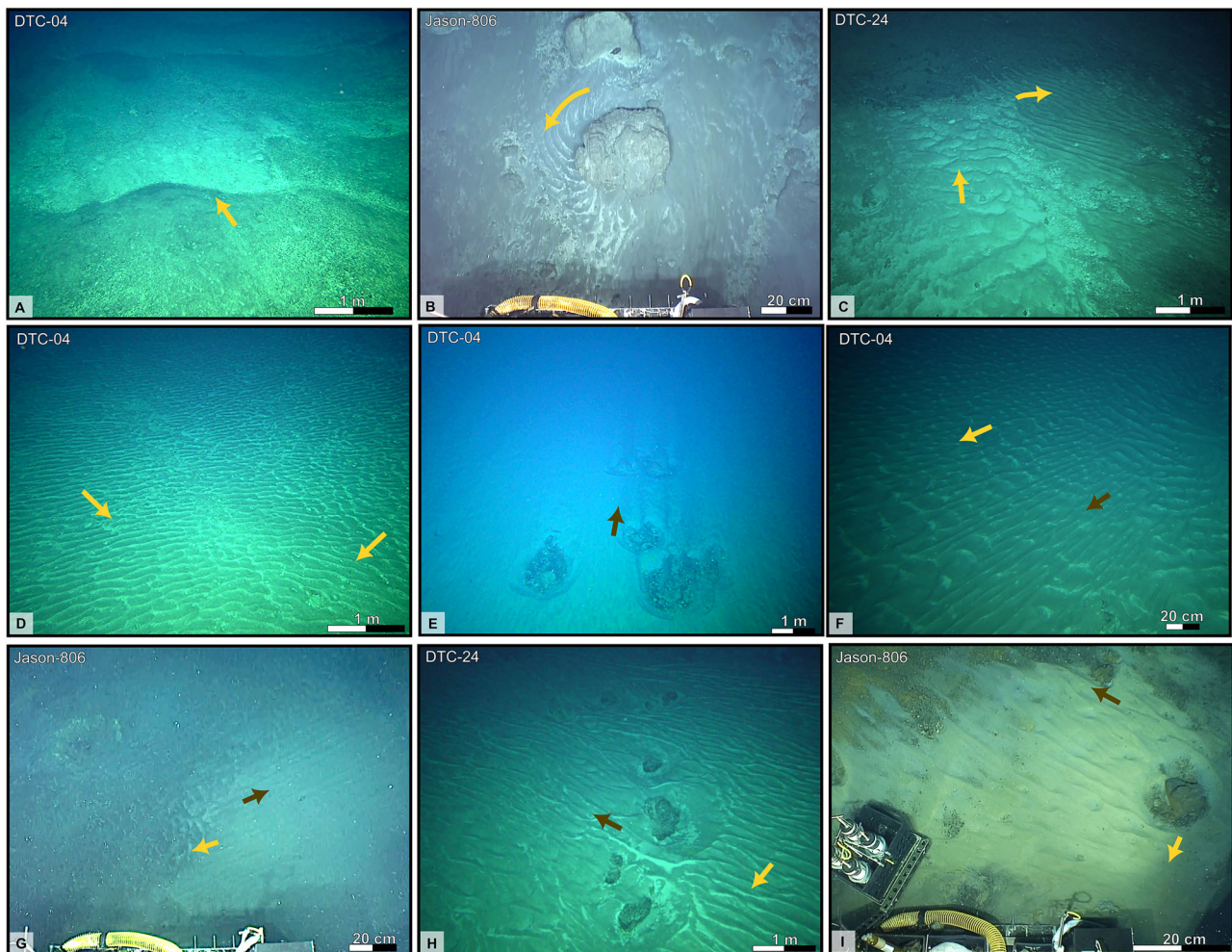


Fig. 3 | Still images of ripples from DTC and ROV videos with interpreted current directions (arrows). Video transect line/ Jason dive number top left. A–C Type 1 ripples or topographically-bound ripples. A Dune bedforms associated with accelerated currents down-slope. **B** Ripple deflection around a large clast. **C** Ripple deflection across a break in slope. **D** Type 2 ripples or slope ripples with conflicting ripple orientations. **E** Obstacle-and-scour marks with interpreted current direction

(dark arrow) **(F–I)** Type 3 ripples or reworked ripples with dark arrow indicating interpreted secondary current direction associated with erosional features. **F** Ripples and sand ribbons with similar interpreted current directions. **G** Ripples and sand ribbons with conflicting interpreted current directions. **H–I** Ripples and obstacle-and-scour marks with conflicting interpreted current directions.

Estimates for the timescales of fluctuation in currents at Havre seamount can be given from the structures observed on the seafloor and approximate current velocities. Bottom-current velocities are estimated from bedform morphology and the bedform-velocity matrix⁴³ (Supplementary Fig. 1). These observations indicate bottom current speeds can approach 1 m/s, but are typically between 0.1–0.5 m/s. Timescales of ripple formation may also be estimated from⁴⁴, where ripple formation in fine sand was demonstrated to reach equilibrium heights and lengths in 1–100 h for currents travelling at 0.25–0.6 m/s. Ripple formation on the seafloor therefor occurs in a matter of hours to days, with changes in velocity likely on a similar timescale. Numerical modelling of currents and in-situ velocity measurements around seamounts from previous studies indicate variations in currents on the order of days^{21–23}, which is consistent with the results given here.

ARGO floats

The global network of ARGO floats provides ocean current velocity data. Data from ARGO floats at 1000 m water depth, within a 60 km radius of Havre Seamount, for the years 2007–2020 (i.e. corresponding to the 2012 eruption date and its aftermaths) were extracted for analysis. Currents up to 1.15 ms^{-1} were recorded west of Havre, whereas weaker currents (ca. 0.2 ms^{-1}) occur to the east (Fig. 1B³⁶). Current direction has no consistent or

uniform orientation over the study area for the time frame selected. Current velocity in the basin west of Havre suggest common flow reversals (between North and South) or migrating, large-scale deep-ocean gyres, which is consistent with observations showing that surface eddies extend to mid-ocean depth (approx. 1000 mbsl) in the Kermadec arc area⁴⁵. Hindcasting of surface ocean currents around Havre Seamount following the 2012 eruption also demonstrated large annual variabilities in current direction⁴⁶. The diversity in current data at 1000 mbsl and at surface around Havre Seamount suggests that larger-scale or more regional topographic effects (i.e. the Kermadec Ridge) may also drive bottom current complexity around Havre Seamount. The ARGO float data demonstrates there is no overarching hydrodynamic regime operating around Havre Seamount in yearly to decadal timescales. As such, the primary hydrodynamic regime around Havre Seamount would be described by fluctuating currents and variability.

Core GC-10

Core GC-10 is a 34 cm gravity core collected from the southern flank of Havre Seamount (Fig. 1B) in a small, perched basin. The upper 23 cm of core is presented in Fig. 5 (full core image provided in Supplementary Fig. 2). The core consists of an oxidized pelagic mud (0–1 cm), a grey, sandy volcaniclastic unit (1–22 cm) and a brown pelagic mud (22–23 cm). The volcaniclastic unit is composed of four distinct sub-units from base to top (1–4),

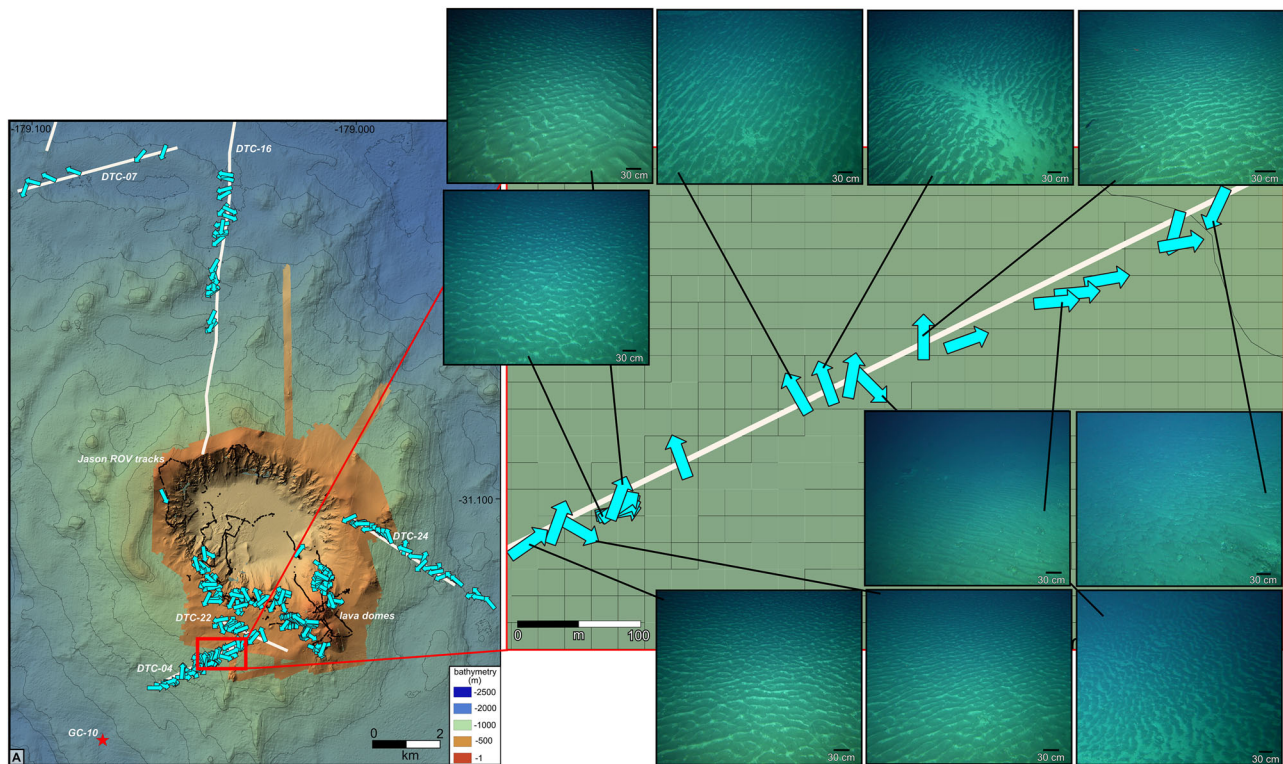


Fig. 4 | Sub-section of Deep-Towed Camera transect DTC-04 showing highly variable ripple orientations over short (m-scale) distances. Total sub-section length ~700 m. Ripple orientation changes from SW (start of sub-section) to East, and then North. Ripple orientation then swings between easterly and northerly for

the remainder of the sub-section. Several ripple morphologies are also visible throughout the section. Background bathymetry for map provided by GEBCO (Derived product from the GEBCO 2021 Grid, made with NaturalEarth by OpenDEM).

starting with a (1) normally graded fine sand to silt (20–22 cm) that is in scoured contact with the underlying pelagic mud. The sub-unit is density graded with a concentration of ferromagnesian minerals at its base and preferential concentration of glass shards at the top. The basal sub-unit is overlain through a sharp and irregular contact by a (2) weakly stratified sand (8–20 cm). This sub-unit has a concentration of ferromagnesian minerals in a concave-shaped lens near its base (18–19 cm). Eleven thin beds of lens-shaped medium-grained sands alternate with fine sand in curved, but conformable contacts, with weak cross-stratification. Above, a sub-unit (3) of normally graded medium sand to silt occurs through a scoured basal contact (4–8 cm). The sub-unit fines upward but is not density graded. The uppermost sub-unit consists of (4) massive fine sand (1–4 cm) that is in conformable contact with the underlying sub-unit (Fig. 5). Chemical composition of tube glass shards in the volcanoclastic unit (Supplementary Fig. 3) match the products of the 2012 Havre eruption^{37,47}. The other glass compositions identified in the volcanoclastic unit are interpreted as products of previous eruptions from Havre Seamount and/or from distant volcanoes.

While mixed compositions and normally graded beds are attributes of turbidites, and stratified sands are characteristic of contourites, the ubiquitous reworking observed on the seafloor around Havre and in the vicinity of core GC-10 strongly suggests that the variable bottom currents are also active in depositing sediment. While turbidity currents are likely to be active on seamount slopes, the overall structures in sub-unit (2) align better with observed structures on the seafloor around Havre and an interpretation of bottom-current reworking. This interpretation is favoured as sub-unit (2) consists of interbedded sands with normal- and reverse-grading, cross laminations, scours, and lenses of ferromagnesian minerals, indicating significant energy fluctuations across the 22 cm of core. This interpretation is supported by the ARGO float data, indicating there are no spatially or temporally persistent bottom currents around Havre seamount that may contribute to contourite formation^{4–6}. Consequently, we interpret the two

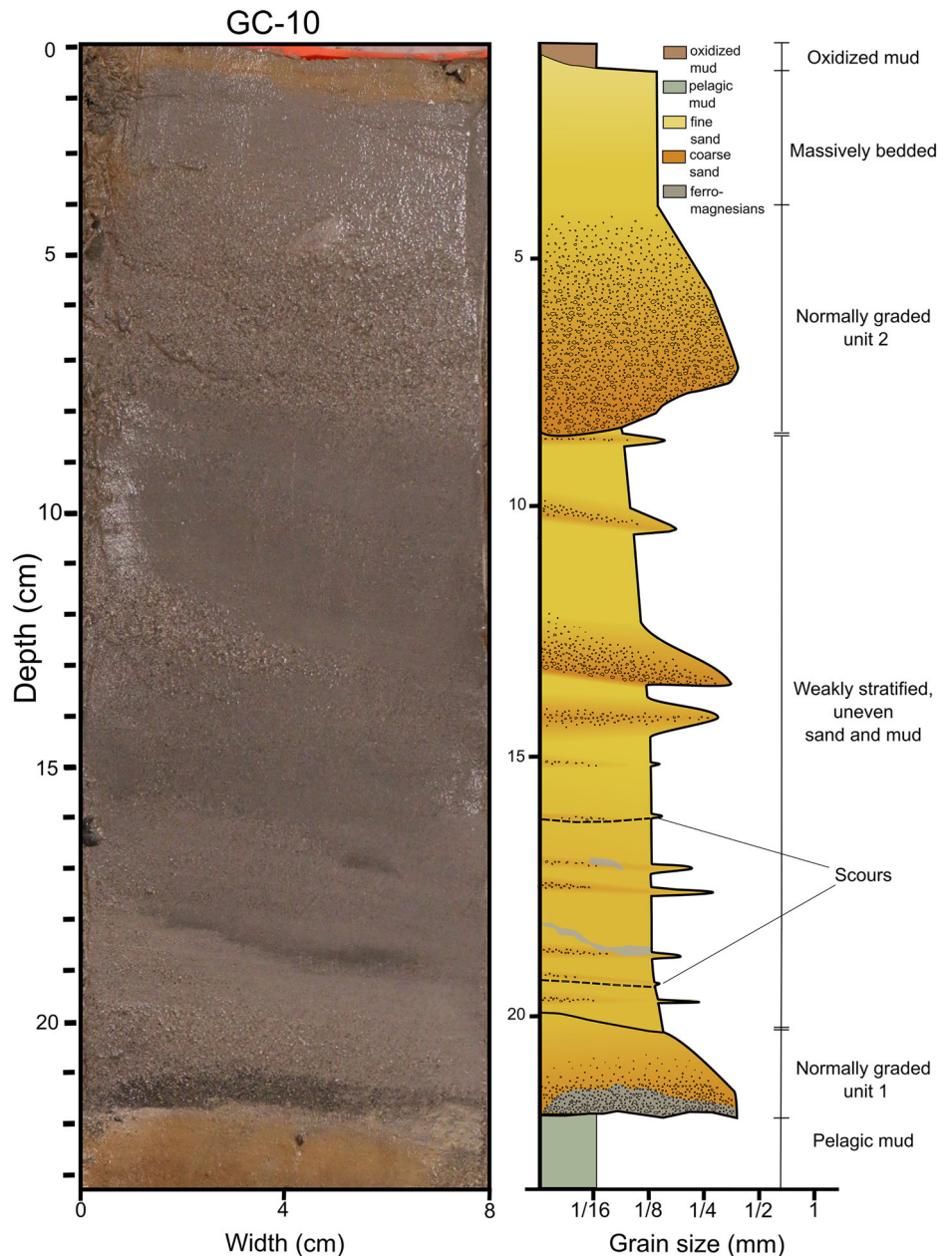
normally graded sands to be thin turbidite beds, and sub-unit (2) to be the product of reworking by variable bottom currents.

Discussion

The seafloor sediments at Havre Seamount are strongly affected by instabilities created from deep- and mid-ocean currents interacting with the edifice. Sediments reveal complex reworking occurs as a result of both spatial and temporal fluctuations in bottom currents on timescales of days. This study shows that currents are enhanced locally, reaching speeds up to 1 ms^{-1} , which is in the same order of magnitude of the speed of a terrestrial meandering river⁴⁸, and is unusual for the mid- to deep-ocean (typical velocities around $1\text{--}2 \text{ cm/s}$ ⁶). We interpret the main factors driving the bottom current variability around Havre Seamount to be 1) large-scale regional and local instabilities that originate from broad topographic features (i.e. the Kermadec Ridge, Havre seamount; Fig. 6A) 2) intermediate-scale (100 s of m) instabilities that arise from bottom currents being steered or altered (i.e. eddying, jets, density gradients; Figs. 6B, C) and 3) small-scale topographic influences such as slope breaks, channels, lava domes and coarse-sized clasts (Fig. 6D). Further, we demonstrate that this variability in bottom-currents exists at a scale of hours/days but is persistent over years.

Although this study is confined to a volcanic intra-arc environment, comparable hydrodynamics can be anticipated where bottom-currents interact with imposing topographic obstacles (m–km, e.g. seamounts, abyssal hills and knolls^{29,49,50}), and experience a degree of instability either from seasonal variability^{51,52} strong eddy effects⁵³, a confluence of currents^{54,55}, or some combination of the above. Remobilized deep-sea sediments with complex erosional and depositional bedforms are therefore likely to be a multi-environment facies that can be identified at other locations globally and in the rock record. To some extent, similar sedimentary features have already been documented in channels, continental margins, and around volcanic edifices^{14,26,27,56}, and intense below-wave base reworking of volcanic sediments has been documented¹⁷. However, these

Fig. 5 | Core GC-10 (left) and strip log (right). Core consists of two normally-graded sand units with a stratified sandy volcanoclastic unit (sub-unit (2)) and a pelagic mud at the top and base of core. Full core description in text.



studies did not present high-resolution data, and/or did not discuss the full effects of such complexity on the seafloor.

The observations from Havre seamount are notable due to the extreme fine-scale variability on both spatial and temporal scales. Further, the sedimentary structures on the seafloor, ARGO float data and core suggest sediment deposition is characterized by variability both in the short-term (days) and long-term (years). This is distinct from deposits of contourites and turbidites, where deposition is defined by either the action of long-duration, persistent currents (e.g. contourites^{4,58,59}), or by episodic down-slope sediment movement (e.g. turbidites^{8,10,11}). While variability can be present in both systems^{27,50,56,60}, it does not represent the defining feature.

We propose a new facies name to accommodate short-term, bottom current-modified deposits of sediment that display high degrees of variability across the entire assemblage. We propose the term *Deep-water Multidirectional Tractional Sands* (DMTS) as a new facies name. The nature of the facies essentially depends on intermediate to small-scale topography and the effects of fluctuating currents, as well as the availability of sufficiently fine sediment for remobilisation. In a sediment-starved setting, DMTS facies will

consist of silt- to coarse-sand sediments dominated by scours and erosive surfaces, with coarse pebble lags (if available) and occasional cross-bedding. Lack of sediment deposition and relatively strong bottom currents may prevent colonization and preservation of bioturbation. In contrast, settings with a supply of sediment that experiences moderate rates of erosion and deposition, DMTS will likely bear a mix of scours, erosional surfaces, pebble lags, and abundant cross-bedding. These sedimentary structures are interbedded, and cross-bed orientation and shape will be variable; benthic life and associated bioturbation may be present^{61,62}. In high sediment accumulation settings, bedding will be dominated by cross stratification of varying orientations and morphologies, planar lamination, and bioturbation. Bioturbation represents may be sporadic throughout the section and depends on additional factors such as nutrient supply. Active submarine silicic volcanoes frequently producing large volumes of fine-grained material, providing an ideal setting for a sustainable development of this facies, however a volcanic origin is not necessary. Given the spatial scale of bed-forms observed and the availability of material to be reworked, DMTS facies may only occur as small, localized patches (m-scale laterally and cm-m

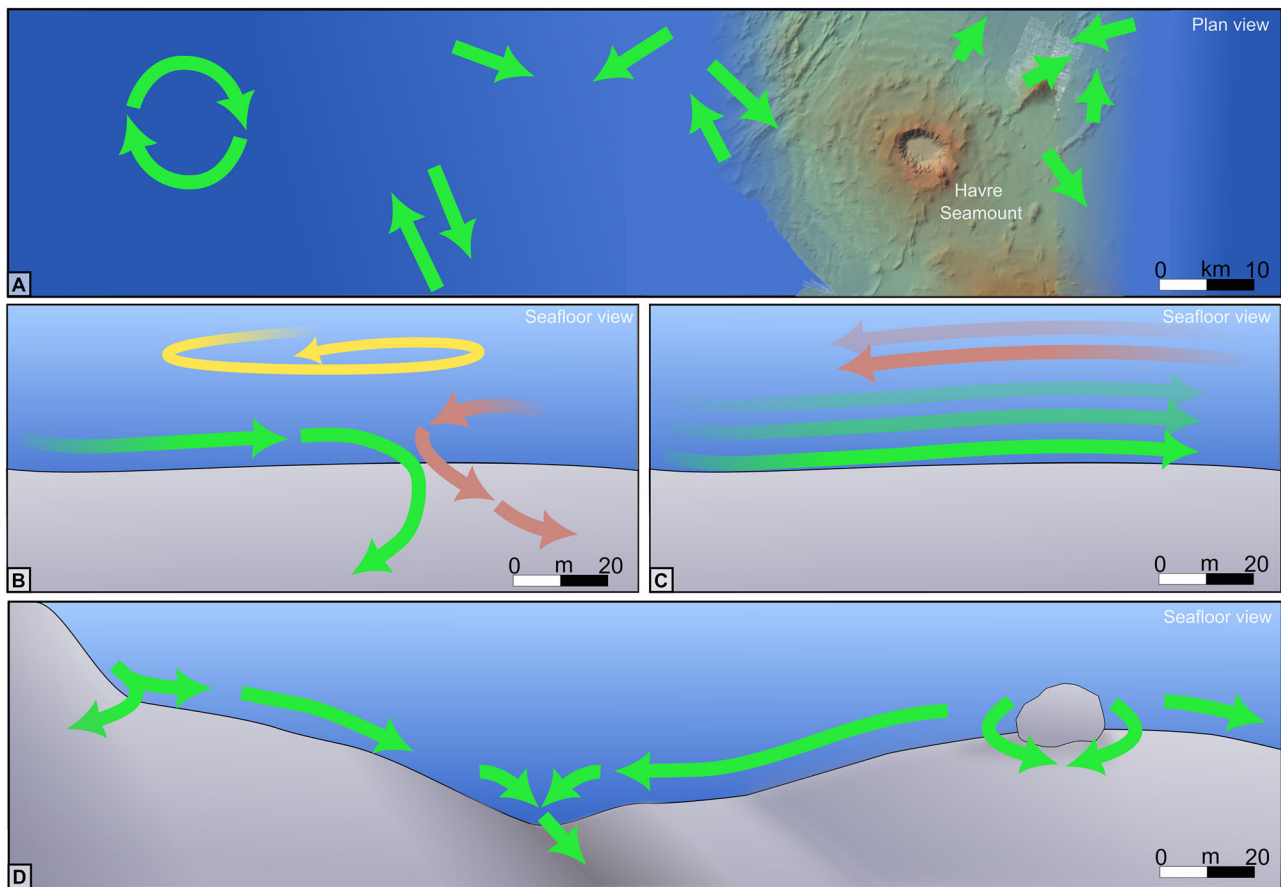


Fig. 6 | Interpretation of the various mechanisms that can affect bottom currents in the vicinity of a seamount. All mechanisms (A–D) may operate simultaneously. **A** Regional current variation over yearly-decadal timescales (modified from ARGO float data, Fig. 1), including regional eddies and directional reverses. **B** Two currents impinging upon each other near the seafloor causing deflection, producing Type 2

ripples (Fig. 3D). Local eddies may also cause current deflection. **C** One bottom current being superseded by another, causing reworking and modification (producing Type 3 ripples, Fig. 3F–I). **D** Current modification by topography (producing Type 1 ripples). Currents may be accelerated downslope, channelized, or deflected around/with topography.

vertically) or as more extensive m- km scale beds up to several m thick. The timescales documented for current fluctuations and ripple formation indicates DMTS facies will be highly chaotic, with frequent changes in grain size, bedform type and bedform orientation.

The fluctuations in bottom currents and sediment assemblages will also likely result in a variability or “patchiness” of the benthic species³⁰, which will be reflected in the bioturbation observed in DMTS. Areas of the seafloor that are frequently disturbed (e.g. by currents, storm events, chemical impacts) can be considered a “disturbance regime” and may create hierarchies of colonization on the seafloor concurrently³⁰. A landscape disturbed by bottom currents may therefore contain a greater biodiversity than environments where uniform conditions persist³¹. Previous mapping of disturbance regimes in benthic habitats has focused primarily on continental margins, where tidal, wave, and storm activity can contribute to disturbances^{30,31}, however this study indicates that disturbance regimes may exist in the open ocean where currents interact with topography and are likely to be common worldwide. As habitat mapping is an important component of ecological management, and the deep-sea is coming under increasing risk from anthropogenic activity, understanding the distribution and nature of these environments is vital.

Here we provide criteria for distinguishing between contourite, turbidite facies models and DMTS (Table 1, Fig. 7). Although there are many possible variabilities in both turbidite and contourite models^{60,63–65} we focus on differentiating from standard models for sandy contourites and turbidites. Of primary significance is scale, as DMTS facies are observed on seamount flanks and summit and generally diminish basin ward deposits are likely to be thin and patchy (m-scale both vertically and horizontally)

although in certain locations may accumulate into deposits 10 s of m thick if the sediment supply allows for it. This contrasts markedly with contourites that may be several km in length and 100 s of m high^{4,66}. Additionally, we can expect lateral connectivity (or a laterally evolving pattern) in turbidite and contourite deposits^{4,10,66,67}, which DMTS facies will not show. Grain size distribution in DMTS is also notably different from contourite and turbidite models. Both turbidites and contourites have overarching grading trends; turbidites should appear normally graded⁸, whereas sandy contourites are characteristically reversely to normally graded¹⁰, each reflecting a consistent change in energy. No such trend will be apparent in DMTS as energy of the current will vary frequently. Further, internal facies of DMTS will not be vertically organized (e.g. Bouma sequences) and should be dominated by cross-stratification of various sizes, angles and directions, with planar lamination, scour surfaces, and pebble lags interbedded sporadically. This contrasts with contourites, which may have an overall massive, bioturbated and mottled character or may have some unidirectional ripple and planar lamination present in the sandy and silty units^{4–6}. Surrounding sediments may also be indicative of deposit type. Stacked sequences of turbidites are common, or turbidite sequences may be encased in pelagic sediments. Contourites may have associated pelagites, turbidites, and possible DMTS facies. DMTS facies may be stacked (and difficult to differentiate) with possible turbidite, contourite and pelagite facies associated. Bioturbation in DMTS will be patchy and dependent on substrate type and energy conditions³⁰.

The data presented here show that the bottom currents around a seamount can be very active in remobilizing sediments on the seafloor. Deposits from turbidity currents, mass wasting events, or volcanic eruptions

Table 1 | Comparison of facies models and common attributes for Turbidites, Contourites and Deep Marine Tractional Sands

	Turbidites	Contourites	DMTS
Depositional setting	abyssal plain (episodic density currents)	adjacent to topography (persistent currents)	adjacent/on topography (variable currents)
Unit scale (horizontal)	m to 1000 s of km	10 s of m to 10 s of km	m to 100 s of m
Unit scale (vertical)	mm to dm	mm to dm	mm to cm
Succession thickness	m to km	m to km	cm to m
Lateral connectivity	excellent	good to moderate	poor
Surrounding sediments	other turbidites, contourites, pelagites	pelagites, slope deposits, DMTS	DMTS, slope deposits, contourites
Overall structure	normally graded	reverse- to normal-grading	stratified
Common facies	Bouma sequence	predominantly massive and bioturbated with mottled appearance, or may include traction structures (low-angle laminations, cross-bedding)	erratic, dominated by traction structures including cross-bedding (low and high-angle) planar lamination, scours, and pebble lags
Bioturbation	uppermost interval only	ubiquitous	patchy

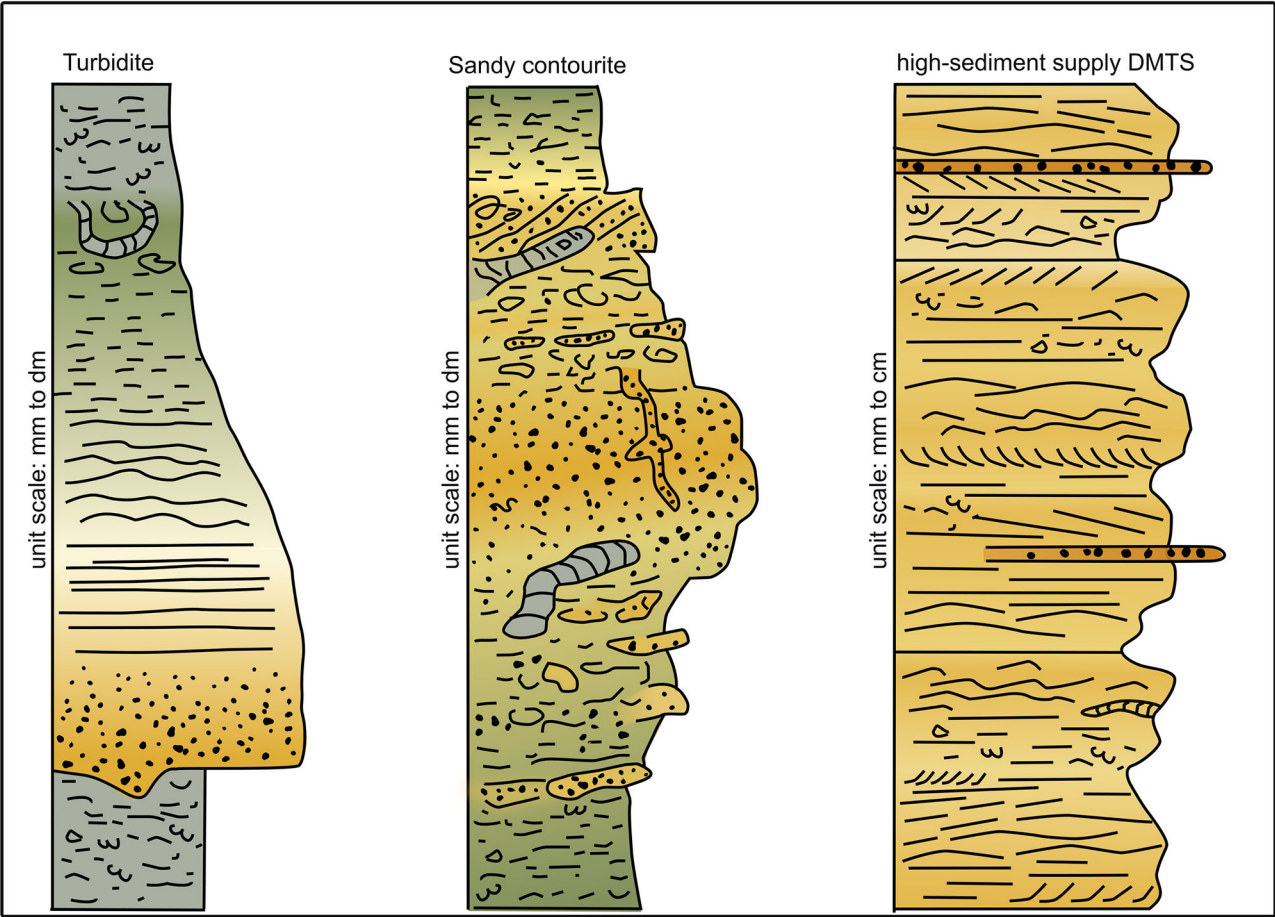


Fig. 7 | Comparison of facies models for turbidites, contourites, and deep marine tractional sands. Modified from ref. 12. Turbidites: Unit scale of mm-dm, will have a characteristic Bouma facies sequence and an overall normally graded structure. Bioturbation may be present in the uppermost facies. Contourites will have a unit scale of mm to dm and will have an overall reverse-to-normally graded structure. Facies may be predominantly massive and bioturbated with a mottled appearance or

may include traction structures (including low-angle laminations, and rare cross-bedding). DMTS will have a unit scale of mm-cm, an overall sandy composition with erratic grain size changes. Facies will be stratified, with various traction structures and paleocurrent directions. Traction structures will be interbedded with scour surfaces and pebble lags. Bioturbation will be patchy.

that are thin and/or fine-grained are unlikely to remain entirely preserved where affected by bottom currents that may rework sediment within months. Data collected in similar settings using photography, video or shallow core in will likely document the effects of bottom currents remobilizing the top (or entirety) of the original deposit, as reworking occurs on

yearly (or much shorter) timescales. Any original feature in areas susceptible to host DMTS will be permanently affected by bottom currents and may obscure the original depositional mechanism. Rapid sediment remobilisation by bottom currents presents a serious challenge for identification of the primary deposits from modern volcanic eruptions or other catastrophes, as

data collection campaigns may take months or years to organize following the event. Importantly, marine tephra layers are classically used to determine magnitude, intensity and volatile budget of past eruptions⁶⁸. In particular, grain size distributions and isopachs in marine tephra layers are used in numerical models to reconstruct eruption dynamics^{68,69}. Our study demonstrates that bottom currents around seamounts are sufficient to winnow and/or remobilize fresh ash, obscuring features that may have existed at the uppermost part of an eruption deposit.

In geology and marine geosciences, current indicators (i.e. ripples, scours) preserved in outcrop and core are a critical tool for interpretation of paleoslope, provenance, depositional environment, morphology of landforms and basin analysis⁷⁰. However, preserved structures offer only a limited subset from the original system and likely result from preservation bias. These points of data are often assumed to be recording uniform hydrodynamic conditions, yet this study demonstrates that ripple and scour marks preserved in outcrop and core may not be representative indicators of a persistent, spatially consistent hydrodynamic regime. The sparse data from commonly poorly laterally continuous outcrops and cores are unlikely to be sufficient to develop a comprehensive or even basic understanding of the local hydrodynamic conditions at the time of deposition where appreciable deep-current/topographic interactions occur. Further, the interpretation of long-term, regional hydrodynamics requires the consideration of the relative short time span required to form underwater topographic features such as seamounts and fault scarps (~10,000 years^{71,72}). It is likely that new topography will develop and drive changes in bottom currents within short depositional timeframes. Additionally, active volcanoes can change small-scale morphologies (i.e. lava domes, channels) within a short (day-month) timespan^{32,73,74}. Thus, variable current directions at a locality may not represent a change in provenance or variations in ocean current, but rather edifice modification through growth or mass wasting.

Dataset and method

Observations were conducted using high-definition videos collected from two independent research voyages to the Kermadec arc/Rangitāhūa. The MESH expedition in 2015 collected 233 hours of video footage within the caldera of the Havre Seamount and on the flanks, using the Remote Operated Vehicle (ROV) Jason 2 as well as a 1-m DEM of the caldera using the AUV Sentry from the RV Roger Revelle⁴². The 2022 VULKA voyage collected seafloor footage along rectilinear transects at and nearby Havre Seamount by Deep-Towed Camera (DTC) from the CSIRO research vessel RV Investigator³⁵. Although the methods of investigation differ, the collected data of the two voyages is scaled, of similar format, is strongly complementary and were both compiled by the first author. We present data from one gravity core, HVR-GC-10 that was collected during the 2015 MESH expedition.

Sedimentary features observed from the video data were described and compiled. Ripple size was estimated using calibrated lasers in the video footage. Active current observations were made by noting the swimming direction of a species of fish remaining stationary against a current just above the seafloor, and by the direction of particle drift. The latitude, longitude and azimuth of each observation was extracted from the metadata of each dive/tow. The locations of 430 discrete video observations were plotted with QGIS 3.16.16 software. Depth and slope values were extracted for each observation from the 1 m resolution DEM Sentry-based bathymetry map³² for most data points, and from the 2022 MBES bathymetry at about 25 m vertical resolution³⁵ for all others. Comparison between the Sentry-based and 2022 MBES bathymetry shows no difference relative to our study at this resolution. Additional large sedimentary features were described using the 1 m DEM. Mid-ocean current velocity data for a 10 km radius around Havre Seamount was extracted from the 800–1200 dbar Scripps Argo trajectory-based velocity product for the date ranges of 2007–2020⁴⁶.

Compositional analyses of volcanic glasses were acquired using a JEOL JXA 8530 F Plus field emission electron microprobe at the Central Science Laboratory, University of Tasmania. The instrument is equipped with a field emission source and an accelerating voltage of 15 kV, a beam current of 10 nA and a beam size of 5 µm was used. The instrument has 5 wavelength

dispersive spectrometers and is operated using the Probe Software Inc. 'Probe For EPMA' software package. Elements were measured using Kα lines and analysing crystals LiFL for Mn, Fe, Cr and V. The elements P, S, Cl, K, Ca and Ti were analysed on PETL using Kα lines and Sr and Ba using La lines. Si, Na, Al and Mg were analysed on TAP using Kα lines and F was analysed on LDE1L using Kα lines. Peak (background) counting times were 10 (10) seconds for Mg, Al, Si, Ca, Ti, Fe, Cl, P, Cr, V and S, 20 (10) s for Na and K, 20 (20) seconds for Mn, Sr and Ba, 30 (30) seconds for F. F was analysed with slope (Lo) to avoid interferences. The elements Al, Si, P, Cl, K, Ca, Cr, Fe and V were analysed with a linear background correction and the elements Na, Mg, Sr, Ti, Mn, Ba and S were analysed with an exponential background correction. Interference corrections for Fe on F, Ba on Ti, Cr on Mn, V on Cr and Ti on V were applied⁷⁵. A time dependant intensity correction was applied to Na, K, Cl and F if applicable. Oxygen was calculated by cation stoichiometry and included in the matrix correction. Oxygen equivalent from halogens (F and Cl), was subtracted in the matrix correction. The matrix correction algorithm utilized was Armstrong/Love Scott⁷⁶ and the mass absorption coefficients dataset was LINEMU⁷⁷ <10 keV and CITZMU >10 keV⁷⁸.

Estimations of current velocities were made using the⁴³ bedform-velocity matrix and observed bedform morphology (Supplementary Fig. 1). Current velocities have been adjusted for volcanoclastic sediments using the hydraulic equivalences established by⁷⁹. The 2012 silicic ash typically consists of very angular particles ranging 16–800 µm⁴¹. The 2012 clastic deposits lay on highly irregular to smooth substrate made of lavas, pumiceous boulders, and fine sediment. For the ash observed at Havre Seamount (16–800 µm) the approximate hydraulic equivalence is fine sand to silt based on a bulk clast density of 445.5 kgm⁻³⁸⁰. The very angular shape of the volcanic particles slightly increases the drag of the particles⁸¹ and thus their equivalent hydraulic size, however this minor effect was not calculated here for simplicity.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

Scripps Argo velocity dataset is freely available via the Argo website (<https://argo.ucsd.edu/data/argo-data-products/velocity-products/>). All other data are available in the main text or the supplementary materials.

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Author contributions

Conceptualization: M.J. Methodology: M.J. and S.F. Investigation: M.J. and S.F. Supervision, M.J. and R.J.C. Writing- original draft: S.F. Visualization: S.F. Writing- review and editing: S.F., M.J., R.J.C. and P.T.H.

Competing interests

The author declare no competing interests.

Additional information

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